

Water Jug Problem

Aim

To solve the Water Jug Problem using Artificial Intelligence state-space search.

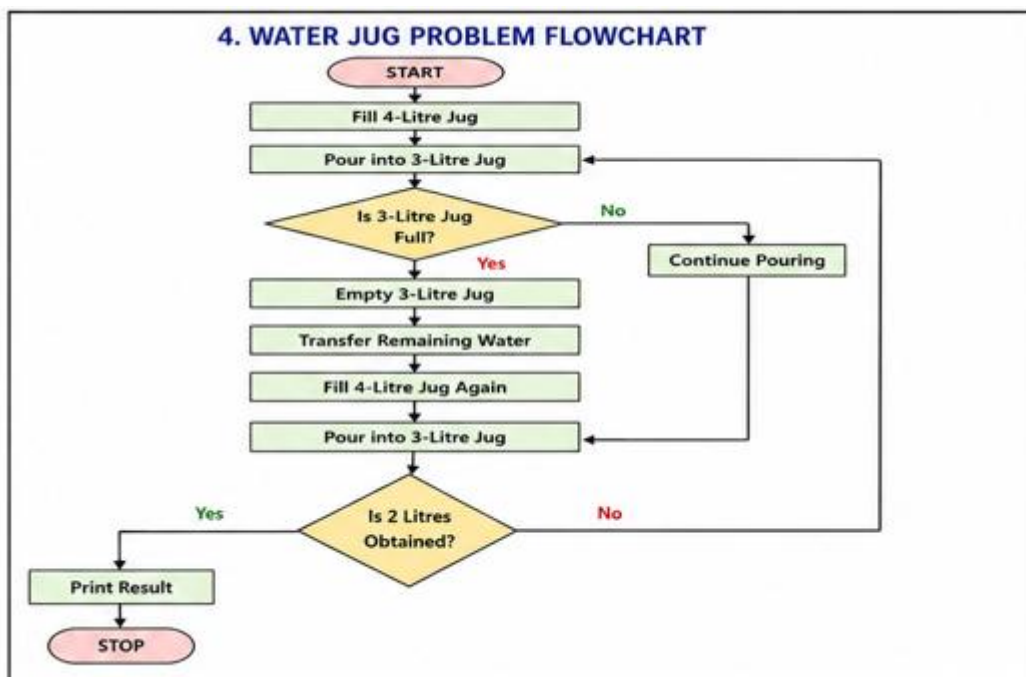
Objective

To obtain the desired quantity of water using two jugs of different capacities.

Algorithm

1. Start with both jugs empty.
2. Fill one jug completely.
3. Transfer water from one jug to another.
4. Empty a jug when necessary.
5. Repeat until the target quantity is obtained.
6. Stop when the goal state is reached.

Flowchart



Sample Output

Water Jug Steps:

4L Jug = 4 3L Jug = 0

4L Jug = 1 3L Jug = 3

4L Jug = 1 3L Jug = 0

4L Jug = 0 3L Jug = 1

4L Jug = 4 3L Jug = 1

4L Jug = 2 3L Jug = 3

Python Code:

```
from collections import deque
```

```
def water_jug(a, b, target):
```

```
    visited = set()
```

```
    queue = deque([(0, 0)])
```

```
    while queue:
```

```
        x, y = queue.popleft()
```

```
        if (x, y) in visited:
```

```
            continue
```

```
        visited.add((x, y))
```

```
        print((x, y))
```

```
        if x == target or y == target:
```

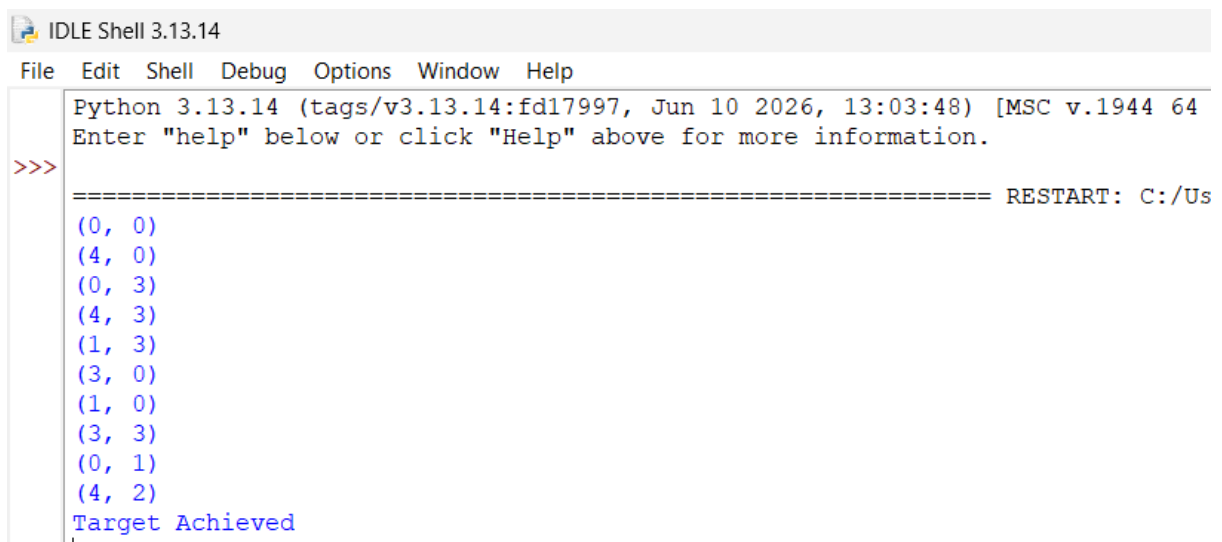
```
            print("Target Achieved")
```

```
            return
```

```
queue.extend([
    (a, y),
    (x, b),
    (0, y),
    (x, 0),
    (min(a, x+y), y-(min(a, x+y)-x)),
    (x-(min(b, x+y)-y), min(b, x+y))
])
```

water_jug(4, 3, 2)

Output:



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Us
(0, 0)
(4, 0)
(0, 3)
(4, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)
Target Achieved
```

Result:

The target quantity of water was successfully obtained.

Conclusion:

The Water Jug Problem was solved using state-space search techniques.